

Q_m with $(Q_m, \psi) =$

$$\begin{aligned} & \Sigma_v (-1)^{d_x+1} \int \dots \{ \int U_{x,y} \partial^{d_x} \psi(F_x(x)) dF_x(x) \\ & + \int F_{x,y}(x, y) (\partial^{d_x})^2 \psi(F_x(x)) U_x dF_x(x) \\ & + \int F_{x,y}(x, y) \partial^{d_x} \psi(F_x(x)) dU_x \} [y \psi'_v(y) + \psi_v(y)] dy_1 \dots dy_{d_y}, \end{aligned}$$

where $U_x, U_{x,y}$ are Brownian bridge processes with dimension $d_x, d_x + 1$, correspondingly; as a generalized random process the limit process Q_m of $\sqrt{n}(\hat{m} - m)$ is Gaussian with mean functional zero and covariance bilinear functional C , given for any ψ_1, ψ_2 by

$$(C, (\psi_1, \psi_2)) = \text{cov}[(Q_m, \psi_1), (Q_m, \psi_2)].$$

Proof. See Appendix.

Similarly to the kernel estimator for the conditional distribution the conditional mean estimator converges at parametric rate as a functional on $D(W)$ for any distribution. When a positive conditioning density exists it is possible to represent the conditional mean as a functional on $D(R^{d_x})$, by the same arguments as in the Lemma. In the case of Remark 3 a similar rescaling provides a faster convergence rate for the estimator considered as a functional on $D(R^{d_x})$.